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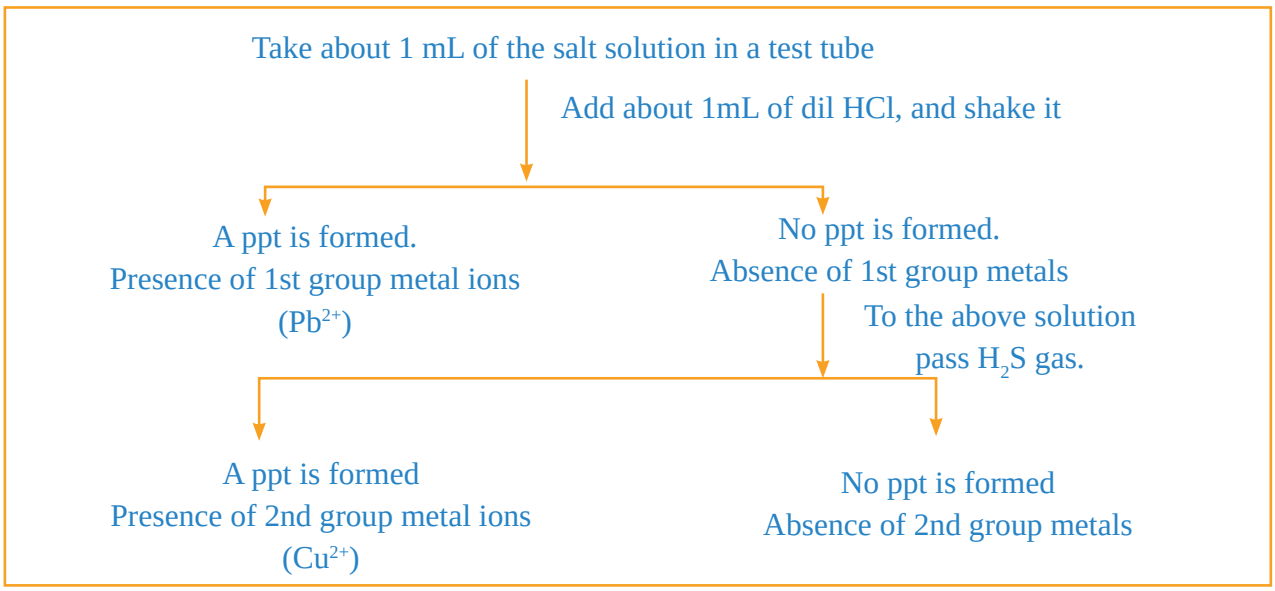
6	Test with barium chloride: Barium forms insoluble ppt with oxalate(BaC_2O_4) and sulphate($BaSO_4$) BaC_2O_4 dissolves in dil. H_2SO_4 and decolourises $KMnO_4$ when gently heated. $BaSO_4$ is insoluble.
7	Test with lead acetate: Lead acetae forms lead sulphate ($PbSO_4$) ppt
8	Brown ring test: The brown ring is due to nitroso ferrous sulphate $[Fe(NO)]SO_4$
9	Neutral ferric chloride test: The reddish brown ppt is due to $[Fe(OH)_2(CH_3COO)]$
10	Ammonium molybdate test: The canary yellow ppt is due to formation of ammonium phospho molybdate $(NH_4)_3[P(Mo_3O_{10})_4]$
11	Sodium nitro bruside test: The purple colour is due to the complex $Na_4[Fe(CN_5)NOS)]$

Preparation of solution of the simple salt for the analysis of cations:

To a small amount of salt in a test tube add 2 to 3mL of water, shake it and gently heat it. If a clear solution is obtained, directly use it for the analysis of cations. If the salt is insoluble, take a small amount of salt in an another test tube, add 3mL of dil. HCl or dil. HNO_3, shake it and gently heat it. If the salt dissolves, use the clear solution for the analysis of cations. This solution is called "original solution".